

Scenario Based Questions

Set Answer 1

Scenario 1: Delivery Time Analysis

Q1. Calculate Q1 and Q3

Logic:

Q1 (First Quartile) represents the 25th percentile of the data, and Q3 (Third Quartile) represents the 75th percentile. They divide the data into four equal parts.

Formula:

$$Q1=(n+1)th/4$$

$$Q3=3(n+1)th/4$$

Q2. Find IQR

Logic:

Interquartile Range (IQR) measures the spread of the middle 50% of the data.

Formula:

$$IQR=Q3-Q1$$

Q3. Detect Outliers

Logic:

Outliers are unusual observations that fall far away from the rest of the data.

Formula:

$$\text{Lower Bound}=Q1-1.5*(IQR)$$

$$\text{Upper Bound}=Q1+1.5*(IQR)$$

Any value outside these limits is an outlier.

Scenario 2: Student Score Analysis

Q1. Mean, Median, Mode

Mean

Logic:

Mean is the average value of all observations.

Formula:

$$\text{Mean} = \sum X/N$$

Median

Logic:

Median is the middle value when data is arranged in ascending order.

Formula:

For odd observations:

$$\text{Median} = (n+1)/2$$

For even observations:

$$\text{Median} = \text{Middle1} + \text{Middle2}/2$$

Mode

Logic:

Mode is the value that occurs most frequently.

Q2. Why Median?

Logic:

Median is less affected by extreme values (outliers), where as the mean can be influenced significantly by very high or very low values.

Scenario 3: Frequency Distribution

Logic:

Frequency distribution shows how many times each value appears in the dataset.

Formula:

Frequency = Count of occurrences

Scenario 4: Multicollinearity

Detect Multicollinearity

Logic:

Multicollinearity occurs when independent variables are highly correlated with each other.

Correlation Method

$$r = \text{Cov}(X, Y) / \sigma_X \sigma_Y$$

VIF Method

$$\text{VIF} = 1 / (1 - R^2)$$

Interpretation:

VIF < 5 → Good

VIF > 5 → Moderate

VIF > 10 → High Multicollinearity

Scenario 5: Medicine Effectiveness Study

Hypothesis Testing

Logic:

Hypothesis testing is used to determine whether the medicine produces a significant effect compared to the place .

Hypotheses

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

(Significant effect)

Test

Use:

Independent t-test

Decision:

$$p\text{-value} < 0.05 \rightarrow \text{Reject } H_0$$

$$p\text{-value} > 0.05 \rightarrow \text{Accept } H_0$$

Scenario 6: Outlier Detection in Sales

IQR Method

$$\text{IQR} = Q3 - Q1$$

$$\text{Lower Bound} = Q1 - 1.5 * (\text{IQR})$$

$$\text{Upper Bound} = Q1 + 1.5 * (\text{IQR})$$

Z-Score Method

Formula:

$$Z = \frac{X - \mu}{\sigma}$$

Interpretation:

$$|Z| > 3$$

Outlier

Scenario 7: Customer Satisfaction

Mean Satisfaction

Formula

$$\text{Mean} = \frac{\sum X}{N}$$

Median Satisfaction

$$\text{Median} = \text{Middle Value}$$

Mode Satisfaction

$$\text{Mode} = \text{Most Frequent Rating}$$

Logic:

Customer satisfaction can be summarized using mean, median, mode, and frequency distribution to understand the overall satisfaction level.